CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Master of Business Administration (MBA)

Post Graduate Diploma in Management (PGDM)

Semester –II University Examination – May 2019

MB747.3 Operations Management (New)

22.05.2019, Wednesday   Time: 10:00 AM to 01:00 PM   Maximum Marks: 70

Instructions:

1. This paper contains six questions.
2. All questions are compulsory.
3. This is a closed book examination.
4. Each of the first five questions will carry 10 marks and the last question will carry 20.
5. Tables / Graphs shall be provided by the institute / university.

Q-1 ***Case-let #1*** **10**

Jiang owned a pizzeria, in Chicago. There were four competing pizza joints within six blocks of his store. The pizza had only served foot traffic since opening its doors 10 years earlier and had been a profitable enterprise; but the sight of another pizza shop opening its doors concerned Jiang. The forth shop appealed to math majors, had cut some of Jiang's business, and the impending opening of Wireless Pizza, a combination Internet café and dining establishment, threatened him even more. For the first time, Jiang was about to introduce pizza delivery. He estimated that half of his customers come from neighborhood. In the summer, he would deliver on bicycles, and he purchased a covered motorcycle for winter deliveries.

What else could Jiang do to create operational advantages to help his business?

Q-2 ***Case-let #2*** **10**

Foxconn Technology Group Chairman Terry Gou said the iPhone will go into mass production in India this year, a shift for the largest assembler of Apple Inc.'s handsets that has long concentrated production in China. Gou said that Prime Minister Narendra Modi has invited him to India as his Taiwanese company plans its expansion in the country. Apple has had older phones produced at a plant in Bangalore for several years, but now will expand manufacturing to more recent models. Bloomberg News reported this month that Foxconn is ready to start trial production of the latest iPhones in the country before it starts full-scale assembly at its factory outside the southern city of Chennai. "In the future we will play a very important role in India's smartphone industry," Gou said at an event in Taiwan. "We have moved our production lines there."

Source : *Mass production of iPhones to start in India, a shift from China, 15 Apr 2019, Bloomberg*

What factors make India an attractive location for manufacturing business? What would be the important factors that iPhone might consider before opening plants in India?

Q-3 ***Case-let #3*** **10**

Till quite recently, Venkat, an operator at Flipkart's sorting centre in Bengaluru, would sort out and assign all packages manually. The task was challenging and time-consuming. The deployment of 100 automated guided vehicles (AGVs) at the centre last month has allowed him to do a lot more with ease. The AGVs can sort and assign packages to the pin codes of the customer just from the encoded information on the package. Flipkart claims it can process up to 4,500 shipments in 60 minutes and, when required, scale up to five times with slight adjustments. "Supply chain for e-commerce is tight. Sorting plays a very critical role from the speed as well as liability perspective, because we have sellers all over the country. Robotics can help us increase the throughput and as it is modular, we can deploy more bots if we need higher throughput during sales days," says Pranav Saxena, head of robotics and automation at Flipkart Internet Pvt. Ltd. Flipkart also has a mobile app for its fulfilment executives, which keeps them informed about all processes. Saxena feels technology-based solutions are very important from both an infrastructure and consumer point of view as they help build trust and allow companies like Flipkart to offer multiple things at scale.

Source : How Flipkart is enhancing its supply chain with robotics, machine learning/ April 14,2019//www.livemint.com/technology/tech-news/how-flipkart-is-enhancing-its-supply-chain-with-robotics-machine-learning-1555261532605.html

Explain the Supply Chain Model of Flipkart. Also explain use of Information Technology in SCM.

Q-4 **(a)** The material D_X is used uniformly throughout the year. The data about annual requirement, ordering cost and holding cost of this material is given below: **05**

Annual requirement: 2,400 units

Ordering cost: $10 per order

Holding cost: $0.30 per unit

Determine the economic order quantity (EOQ) of material D_X using above data.

(b) A refrigerator supplier has experienced the following demand for refrigerator during past five months. **05**

Month	Demand
February	20
March	30
April	40
May	60
June	45

Find out the demand forecast for the month of July using five-period moving average & three-period moving average using simple moving average method.

Q-5 System Software (P) Ltd. is planning to develop new software. It has identified the activities given in the table below for this project, and has estimated three time estimates (in days) for every activity. **10**

Activity	Description	Predecessor(s)	a	m	b
A	Plan the objectives of the project	--	1	2	3
B	Choose the appropriate operating system	A	3	5	7
C	Generate Algorithm	A	6	10	14
D	Choose suitable programming language (C++, JAVA/ Basics)	A	4	6	8
E	Write the programme	B,C,D	8	9	10
F	Test the Software	E	2	4	6
G	Get the approval for commercialization	F	1	3	5

Based on the above information calculate the following:

i. Prepare a Gantt Chart covering all the activities.
ii. Draw a network diagram corresponding to this project.
iii. Obtain the ES, EF, LS and LF of all the activities and determine the critical path.
iv. Calculate total float time of the network.
v. Find expected duration of the project and the probability that the project will be completed within 30 days.

Q-6 ***Case Study #***

It is 7:00 a.m. and the siren sounds high at Kandivli (a suburb of North Mumbai) plant of Mahindra & Mahindra's (M&M) tractor division, signaling the starting time of the morning shift. Hardly any workers have turned up. Reporting late on duty is a norm for the workers here. Seldom does the morning shift start before 7:30 a.m. During the day shift, it was an ominous scene to find workers stretching out under the trees and relaxing during the working hours. †he union leaders hung around the factory without doing any work at all. A few days back, the workers in the night shift had beaten up a milkman for creating a lot of noise in the wee hours of the morning and thus, disturbing their sleep during their working hours. Things were worse at the other plant of M&M in Nagpur. But this was all in the 1980s. M&M has come a long way since then—it has won the most coveted Deming prize for quality, and started a farming equipment assembling plant in the USA. After the huge success there, the company opened a second assembly plant and a distribution center in Georgia. Now, the company is in the process of establishing assembling units in Canada to locally produce and market a range of low horsepower cab tractors with features such as AC heater (keeping in view the cold weather conditions for the farmers there), personal stereo, and even a sun roof.

It has also acquired Jangling Tractors in China, which it would use to develop low cost products suited to plough deeper into the US farm equipment market. Now, the fourth-largest tractor company in the world, M&M, has four tractor plants in India (Mumbai, Nagpur, Rudrapur in Uttranchal, and Jaipur). It has been maintaining its market leadership for the past two decades. During the late 1980s, the company tried to apply TQM concepts such as quality circles without getting any success. M&M was the market leader in the tractors segment at that time, but in view of the looming multinational threat in the near future, its internal situation was very fragile. During 1990–94, the company started the use of statistical process control and tried to perform business process reengineering. Its journey towards the Deming prize was initiated in 1994, with the appointment of Prof. Fasutoshi Washio, a Japanese expert, in the implementation of the Deming guidelines. The same year, the company was rechristened M&M Farm Equipment Sector (FES).

Initially, Prof. Washio was skeptical about the Indian companies and workers. He felt that the Indian companies are more like the American companies, which feel that results are important. On the other hand, for the Japanese, the process is more important. Moreover, he had serious doubts about the attitude of the Indian workers with respect to teamwork—a Deming prerequisite—as he felt that Indians were individualistic. He was proved wrong by the M&M workers. In his own words, The Indians can be good team workers, much better than the young in Japan today and, in that sense, perhaps, Deming is better suited to Indian companies'. In the initial few years of interaction with the management of FES, Washio found himself isolated due to disagreements on various fronts. Washio had major difficulties in making most of the Indian companies understand the importance of implementation over creating a perfect framework. In his own words, Indians are very good with framework and the big picture, but are poor with implementation. The kaizen is weak.' Kaizen means gradual, orderly, and continuous improvement in work processes. It took a while for Washio to make the FES personnel understand that good kaizen hinges on implementation, so there is no need to spend too much time creating a perfect framework. Once you start implementing these, the rest will happen automatically. The FES created a team to implement the Deming guidelines. The team identified eleven key areas to be fulfilled:

1. top management leadership and involvement
2. creating and maintaining TQM frameworks
3. quality assurance
4. management systems
5. human resource development
6. effective utilization of resources
7. understanding TQM concepts and values
8. use of scientific methods
9. organizational power
10. relationship with stakeholders
11. enabling of unique TQM activities

In addition, there is another Deming must-do: eliminate dependence on inspection to achieve quality by building quality into the product in the first place. The system at FES earlier was that at the end of the assembly process or at the customer's place, there used to be a final inspection. If a product showed serious flaws then, it was sent again to the shop floor. †his wasted a lot of time and effort, and it did not add to the improvement in the quality of the manufactured product. In order to change this system, computers were installed on the shop floor for showing the standard operating procedure (SOP) of a particular process to make the workers understand the various steps in a process. This reduces the chances of human error and acts as a natural check. At the end of every complete process, a check is performed by a trained worker, who also follows an SOP. Employee involvement is the first step in ensuring the success of any quality initiative. At FES, the workers would dictate terms to the shift supervisors by saying that they would not do different tasks on many machines. The management took time to convince them by giving them examples such as: if your wife can do multiple tasks of cleaning the house, feeding the children, and washing the clothes, why can't you do the same? The workers were explained the multinational threats looming large.

They were told that, if they did not mend their ways, the company might shut down the factory, or even worse, a multinational may take it over and would invariably lay off all the problem-creating workers. Examples of companies shut down in Mumbai due to the changed scenario were given. †he entire programme was termed Ashwamedh' and analogies were drawn from mythology and the current competitive situation. †his brought a complete transformation in the workforce that was now willing to perform multiple tasks, double their productivity, and maintain shift discipline by reporting on time. The workers were informed by the management about every difficulty faced by the company in beating the competition in the marketplace. Some of the workers were sent with the marketing staff to meet the farmers using the company's products and facing problems. †his was called Operation Hamla'.The workers came back chastised and sobered when they realized that a small mistake on the shop floor could cost a farmer his season's crop. †he company even sent some of the union leaders for short training courses in the USA and UK.

This sustained effort on part of the company has paid rich dividends. Costs are down by 15% and the market share has risen by one per cent to 27.3% (10% higher than its closest competitor), despite an overall decline in the tractor demands. The break- even point for a new model of a tractor has decreased to 30,000–32,000 from the 54,000 tractors three years ago. †he worker productivity levels have increased by 100%. Tractor exports from the company have increased 100% over the past 10 years, with 70% to the USA alone. The quality of tractors has improved drastically with the number of complaints per 1000 tractors dropping from 228 to 90. The rejection rate for components bought from vendors, rejection and rework in machining, and rejection at final testing have all been brought down to near zero levels. FES has introduced 15 new models in accordance with the requirements in the international markets. The journey to world- class quality is not over yet. The company now aims at matching the world benchmarks in productivity and quality to establish a cost leadership in the Indian industry.

Based on the above Case Study answer the following:

(a) If you were a part of the top management at M&M FES, how would you have involved the workers in the Deming programme? **10**

(b) Do you think that M&M FES has a strategic quality management system in place? Justify your stand. **06**

(c) How important is the role of top leadership in the success of a quality improvement programme? Discuss in the context of M&M FES. **04**